
Apply now!
Application deadline
for TUD students:
May 8, 2026
24 Aug | 28 Aug
EUTOPIA Summer School
AI in Business
and Healthcare
tud.link/ed5d81
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Ca' Foscari University of Venice
Dresden University of Technology
EUTOPIA
Co-funded by the European Union

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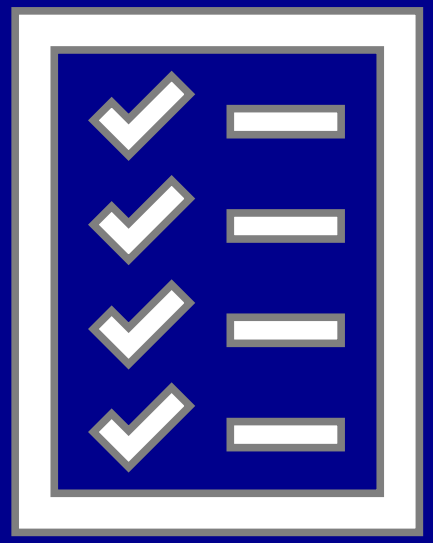
Patrick Zschech
Professorship for Intelligent Systems and Services

Gestaltungsansätze der Advanced Data Analytics

Kick-off Meeting
Summer Term 2026

INTELLIGENT INFORMATION SYSTEMS


Attention!
This course is about you and
your individual development



1) **Lecturers' background**
2) Students' background
3) Motivation for this course
4) Organizational matters
5) Choice of project topics

Patrick‘s background in a nutshell



White-Box AI

Our junior researcher group (previously @ FAU)





Our research focus

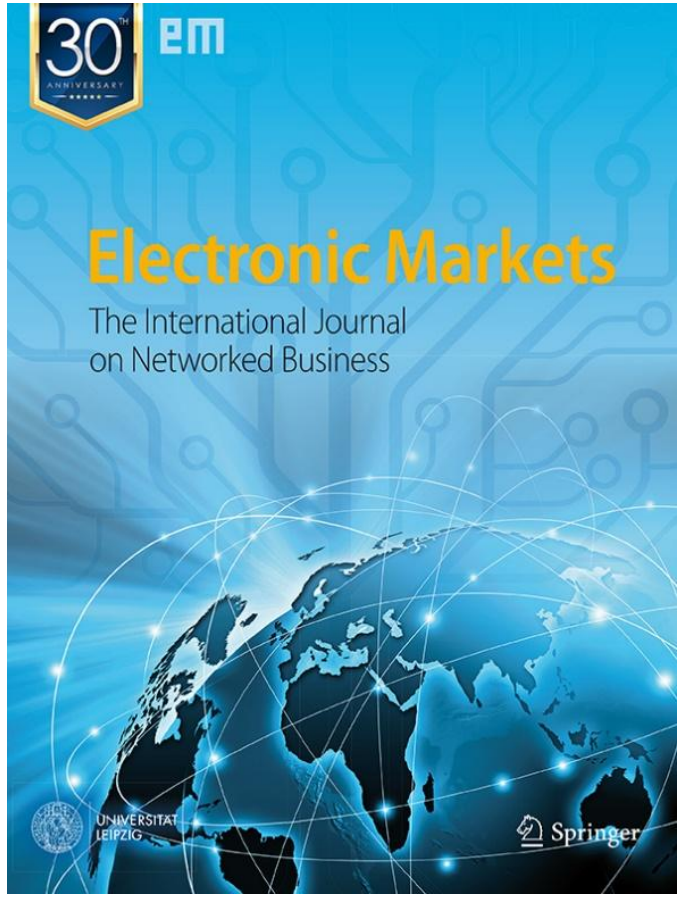
- **Design of intelligent systems:**
  Socio-technical design of intelligent information systems based on modern data science technologies (e.g., deep learning, generative AI, process mining, natural language processing, computer vision)

- **Explainable artificial intelligence:**
  Development of algorithmic innovations in the field of intrinsically interpretable AI and machine learning models (e.g., generalized additive models, optimal decision trees) as well as post-hoc analytical explanation methods for black-box models (e.g., based on SHAP or LIME)

- **Applied machine learning:**
  Application and benchmarking of modern machine learning algorithms in a wide range of application domains (e.g., healthcare, manufacturing, finance)

- **Human-AI interaction:**
  Empirical investigation of human perception and use of AI systems through user-centered experimental studies (e.g., to investigate algorithm aversion vs. algorithm appreciation)

Machine learning and deep learning
Some recommended readings for interested readers ;-)



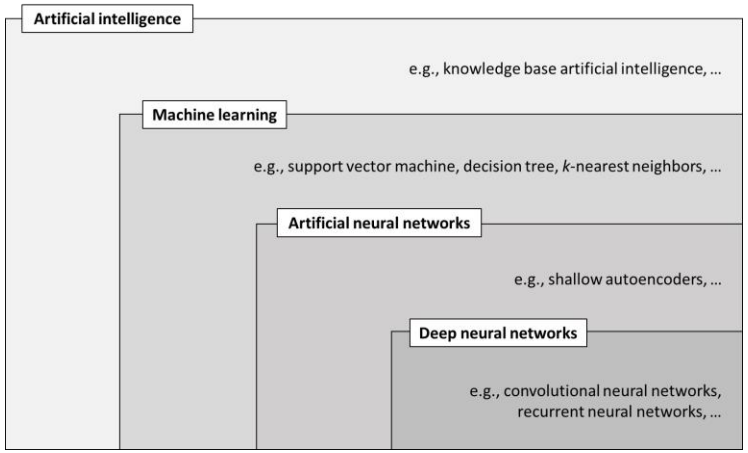
Electronic Markets (2021) 31:685–695
https://doi.org/10.1007/s12525-021-00475-2

FUNDAMENTALS

Machine learning and deep learning

Christian Janiesch[1] · Patrick Zschech[2] · Kai Heinrich[3]

Received: 7 October 2020 / Accepted: 19 March 2021 / Published online: 8 April 2021
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Abstract

Today, intelligent systems that offer artificial intelligence capabilities often rely on machine learning. Machine learning describes the capacity of systems to learn from problem-specific training data to automate the process of analytical model building and solve associated tasks. Deep learning is a machine learning concept based on artificial neural networks. For many applications, deep learning models outperform shallow machine learning models and traditional data analysis approaches. In this article, we summarize the fundamentals of machine learning and deep learning to generate a broader understanding of the methodical underpinning of current intelligent systems. In particular, we provide a conceptual distinction between relevant terms and concepts, explain the process of automated analytical model building through machine learning and deep learning, and discuss the challenges that arise when implementing such intelligent systems in the field of electronic markets and networked business. These naturally go beyond technological aspects and highlight issues in human-machine interaction and artificial intelligence servitization.

Janiesch, C., Zschech, P., & Heinrich, K. (2021) Machine Learning and Deep Learning. *Electronic Markets*. (https://doi.org/10.1007/s12525-021-00475-2).

Generative artificial intelligence

Some recommended readings for interested readers ;-)



Bus Inf Syst Eng
https://doi.org/10.1007/s12599-023-00834-7

CATCHWORD

Generative AI

Stefan Feuerriegel · Jochen Hartmann · Christian Janiesch · Patrick Zschech

Output Modality (Selection)	Model level: *Underlying AI model for different data modalities (e.g., image, text, code)*	System level: *Embedding model functionality to provide interface for interaction*	Application level: *Solving dedicated business problems and stakeholder needs*
Text Generation	X-to-text models, e.g., GPT-4 and LLaMA 2	Conversational agents and search engines, e.g., ChatGPT and YouChat	• Content generation (e.g., SEO and customer service) • Translation and text summarization
Image/Video Generation	X-to-image models, e.g., Stable Diffusion and DALL-E 2	Image/video generation systems and bots, e.g., Runway and Midjourney	• Synthetic product and advertising visuals • Educational content
Speech/Music Generation	X-to-music/speech models, e.g., MusicLM and VALL-E	Speech generation systems, e.g., ElevenLabs	• AI music generation • Text-to-speech generation (e.g., news, product tutorials, etc.)
Code Generation	X-to-code models, e.g., Codex and AlphaCode	Programming code generation systems, e.g., GitHub Copilot	• Software development • Code synthesis, review, and documentation

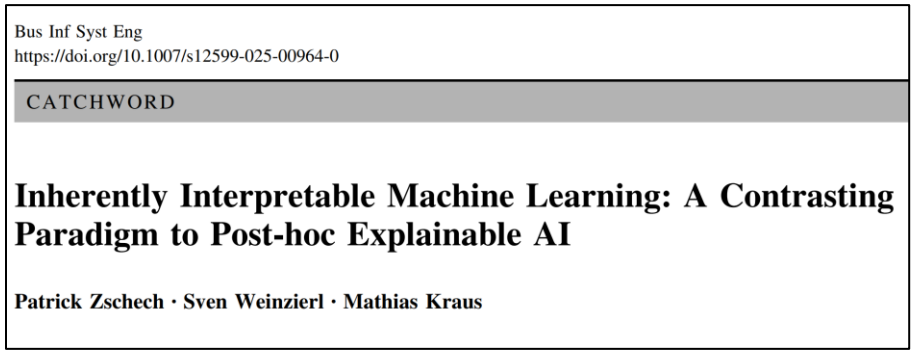
The term *generative AI* refers to computational techniques that are capable of generating seemingly new, meaningful content such as text, images, or audio from training data. The widespread diffusion of this technology with examples such as Dall-E 2, GPT-4, and Copilot is currently revolutionizing the way we work and communicate with each other. Generative AI systems can not only be used for artistic purposes to create new text mimicking writers or new images mimicking illustrators, but they can and will assist humans as intelligent question-answering systems. Here, applications include information technology (IT) help desks where generative AI supports transitional knowledge work tasks and mundane needs such as cooking recipes and medical advice. Industry reports suggest that generative AI could raise global gross domestic product (GDP) by 7% and replace 300 million jobs of knowledge workers (Goldman Sachs 2023). Undoubtedly, this has drastic implications not only for the Business & Information Systems Engineering (BISE) community, where we will face revolutionary opportunities, but also challenges and risks that we need to tackle and manage to steer the technology and its use in a responsible and sustainable direction.

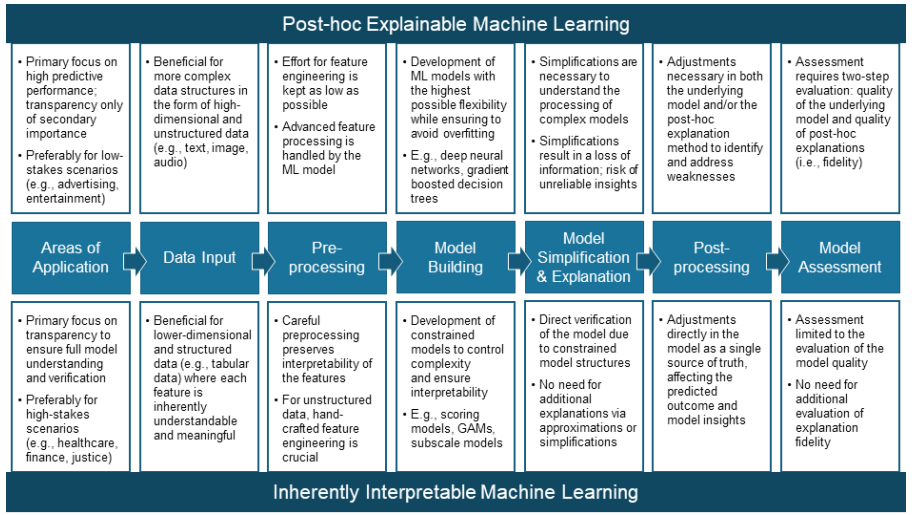
Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business & Information Systems Engineering*. (https://doi.org/10.1007/s12599-023-00834-7)

Inherently interpretable machine learning

Some recommended readings for interested readers ;-)







A decade ago, data scientists devoted roughly 60–80% of their effort to manual feature engineering, aiming to create features with high predictive power and meaningful properties (Press 2016). However, with the rise of deep learning and highly flexible models capable of processing raw input features and learning intricate feature representations, this focus has shifted dramatically (Janiesch et al. 2021). Today, data scientists spend much of their time explaining opaque machine learning (ML) models because users struggle to understand their complex decision logic (Bauer et al. 2021).

As ML models have become integral to data-driven decision-making across many disciplines, including Business and Information Systems Engineering (BISE) and Information Systems (IS) (Abdel-Karim et al. 2021; Duda et al. 2024), the need for model comprehensibility is more critical than ever (Bauer et al. 2021; Brasse et al. 2023). That is, human users must be able to understand the process of how an ML model comes from input data to its predictions based on its internal structure and the relationships it has learned. This property is critical not only for building confidence in ML models for high-stakes domains like healthcare and finance, but also for debugging flawed model structures, supporting regulatory and ethical compliance, enabling users to learn from ML insights, and fostering broader and more responsible adoption in business environments.

Zschech, P., Weinzierl, S., & Kraus, M. (2025). Inherently Interpretable Machine Learning: A Contrasting Paradigm to Post-hoc Explainable AI. *Bus Inf Syst Eng*. (https://doi.org/10.1007/s12599-025-00964-0)

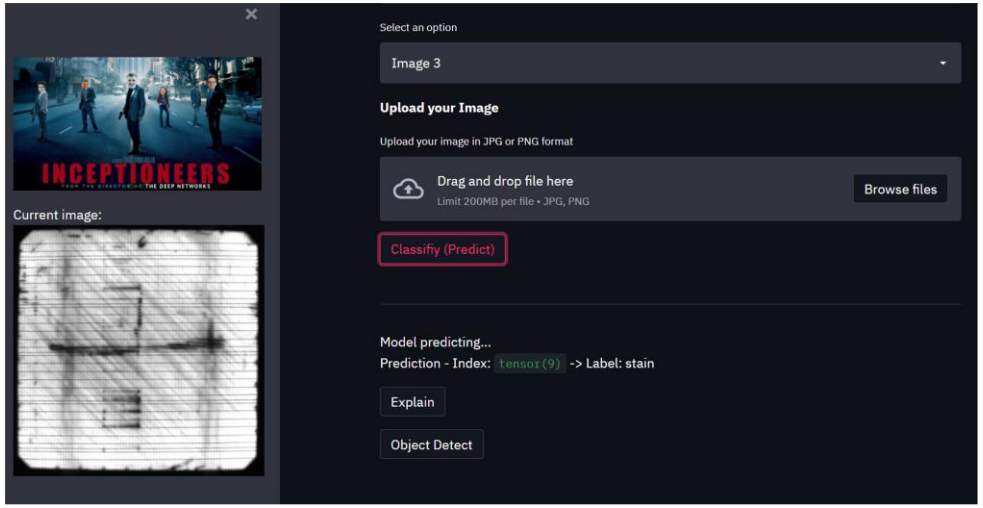
Our teaching portfolio

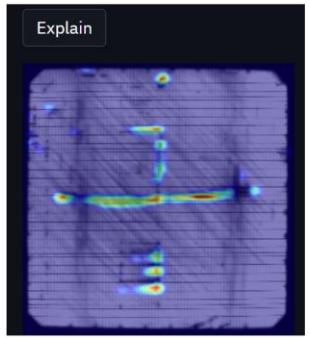
- **Entry level (Bachelor + Grundstudium)**
  - (Einführung in die Wirtschaftsinformatik)
  - **Programmierung und Datenbanken**
  - **Machine Learning for Business Analytics**
  - **Process Analytics**
  - (Einführung in wissenschaftliches Arbeiten)

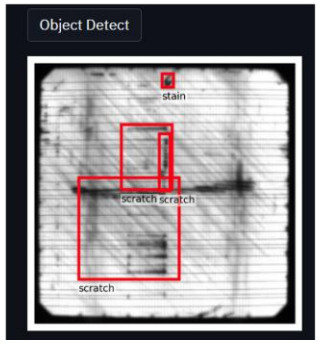
- **Advanced level (Master + Hauptstudium)**
  - **Artificial Intelligence for Advanced Analytics**
  - **Gestaltungsansätze der Advanced Data Analytics**
  - Forschungsbeleg Wirtschaftsinformatik / Belegarbeit
  - Forschungsseminar / Forschungsdesign Wirtschaftsinformatik

Some results from our students
Working closely together with industry

Quality control of solar panels



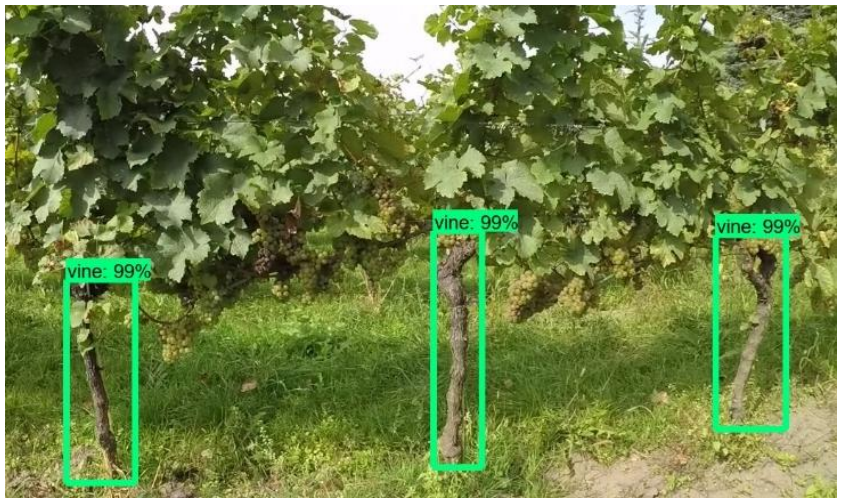


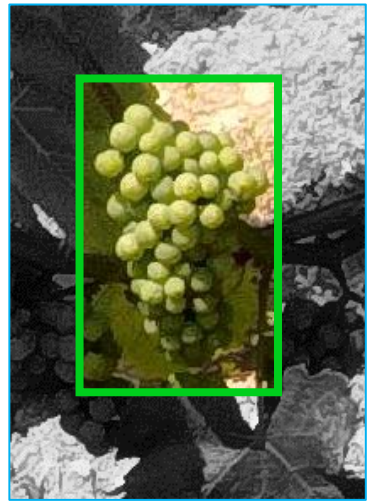


(Source: Harl et al. 2021)

Counting of vine plants, grapes and berries for yield estimation



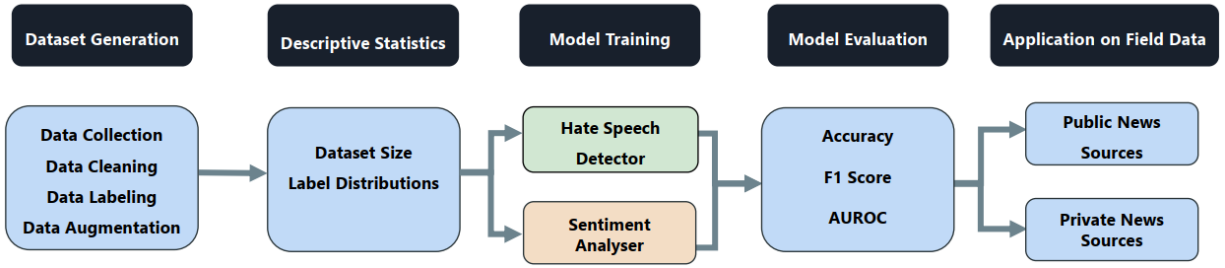




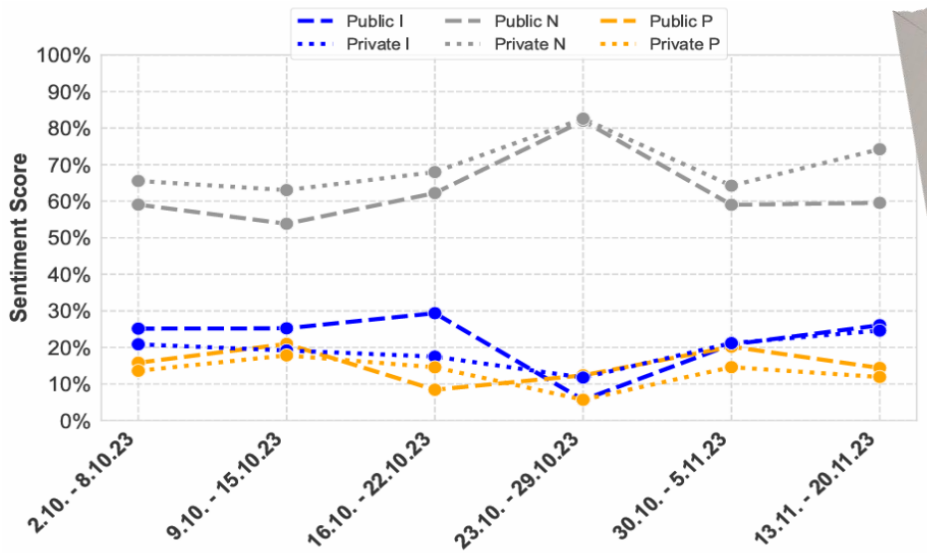
(Source: Heinrich et al. 2019)

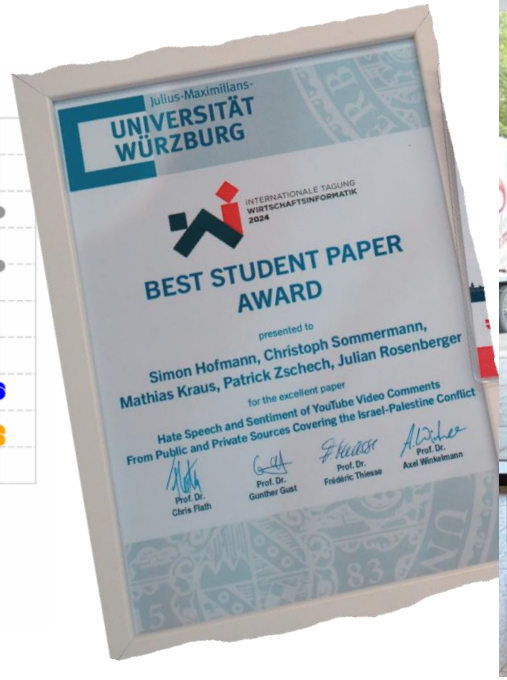
Some results from our students

Best Student Paper Award @ WI 2024 in Würzburg



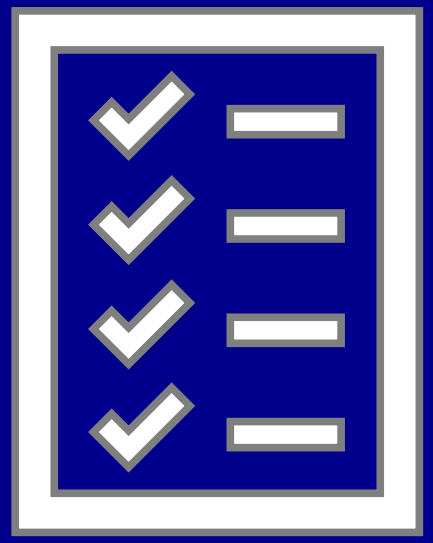
Hate Speech and Sentiment of YouTube Video Comments From Public and Private Sources Covering the Israel-Palestine Conflict







Hofmann, S., Sommermann, C., Kraus, M., Zschech, P., & Rosenberger, J. (2024). Hate Speech and Sentiment of YouTube Video Comments From Public and Private Sources Covering the Israel-Palestine Conflict. *Proceedings of the 19th International Conference on Wirtschaftsinformatik (WI)*. https://aisel.aisnet.org/wi2024/82

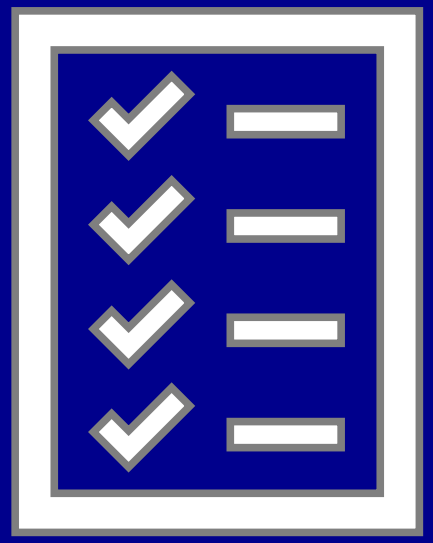


1) Lecturers’ background
2) **Students’ background**
3) Motivation for this course
4) Organizational matters
5) Choice of project topics

What's your background?

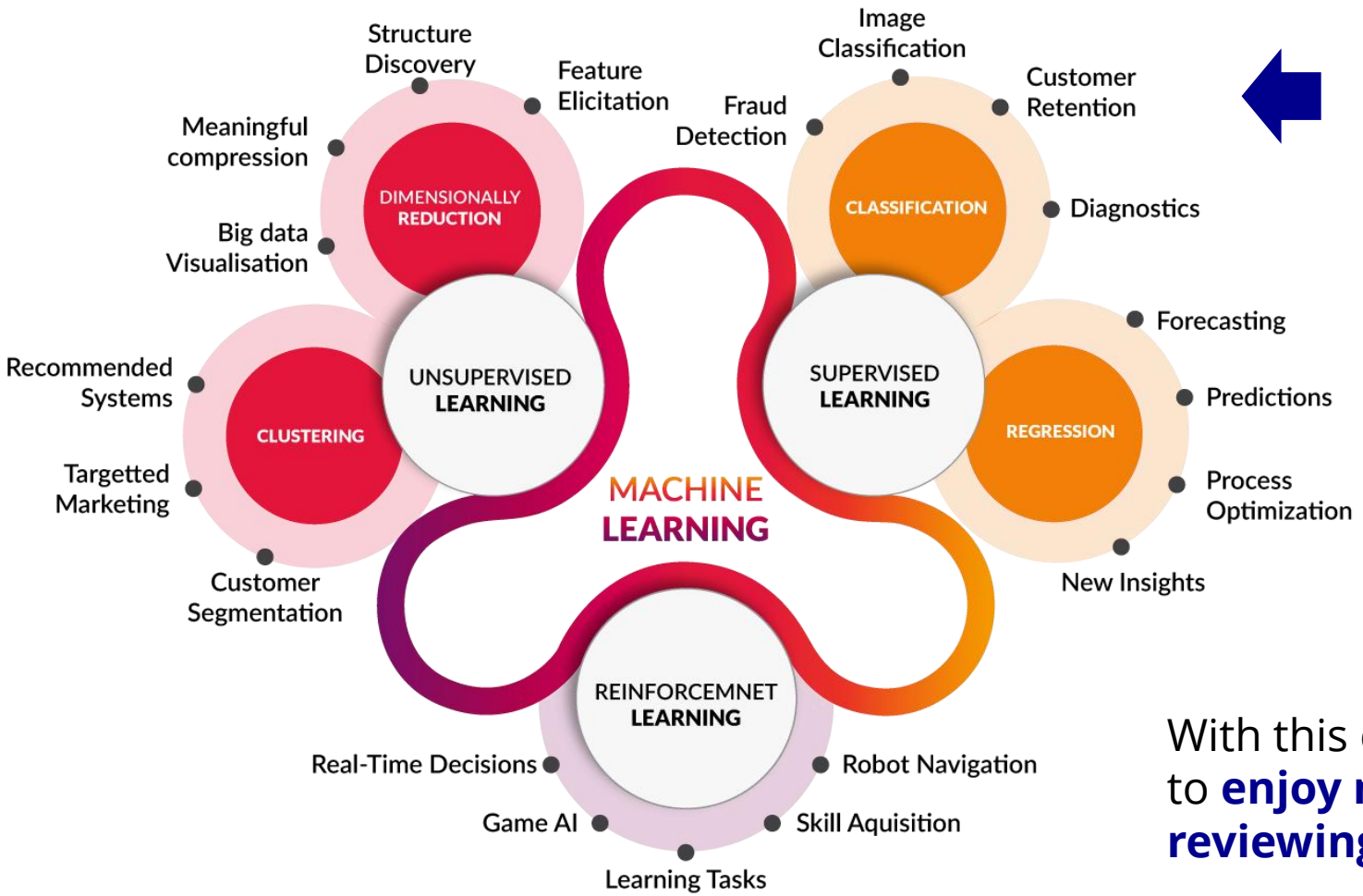


https://images.app.goo.gl/me5mv1[illegible]3bdSHzqc76



1) Lecturers' background
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Motivation for this course



There are so many things we need/want to learn in this field that simply cannot be covered in individual lectures/exercises.

The best way to learn **data science**, **machine learning**, and **AI** is to practically apply these approaches to real-world challenges and derive your own insights.

With this course, we want to give you the opportunity to **enjoy maximum freedom** while regularly **reviewing your progress** on a particular task.

Motivation and course description I

- In an increasingly data-driven economy, the ability to analyze complex data effectively and translate it into informed decisions is a decisive competitive advantage.

- This course combines modern approaches from the field of **advanced data analytics** with practical problems from business information systems and analytics consulting. You will learn to systematically analyze data-driven problems, select appropriate methods, and design innovative solutions.

- The focus is on independently developing the competencies and skills required to successfully complete the project. You will set your own priorities, delve deeper into specific methods and tools, and further **develop your personal competency profile** in line with your interests and goals.

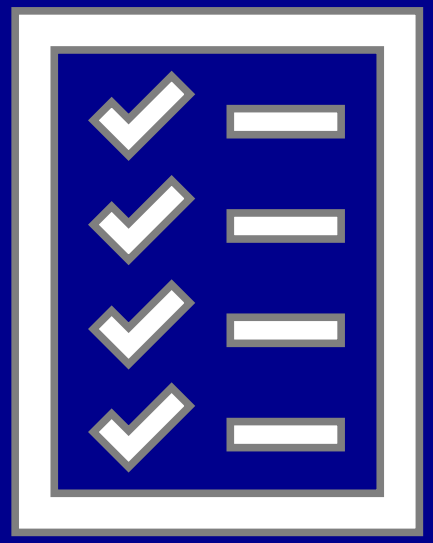
- To best support this individual development, **traditional lecture or exercise formats are deliberately avoided**. Instead, the project offers a great deal of creative freedom, allowing you to explore your own solutions, try out new approaches, and develop in different directions.

Motivation and course description II

- As part of a **project**, you will apply your knowledge directly, work in a **team** on **real-world scenarios**, and present your results in a way tailored to your audience. Whether you want to deepen your analytical skills or actively shape data-driven projects, this module provides you with the foundation to create real value from data.

- *The course builds on existing knowledge and initial practical experience with **data science methods**, **machine learning** and **AI techniques**, as well as **business intelligence tools** and other analytics approaches.*

- The relevant fundamentals can be acquired, for example, in the following courses:
  - Programmierung und Datenbanken (PuD)
  - Machine Learning for Business Analytics (ML4BA)
  - Artificial Intelligence for Advanced Analytics (AI4AA)



1) Lecturers' background
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Project timeline and key dates

Kick-off meeting & topic overview

- **Date:** Wednesday, **April 15**, 2026, time: 11:10 – 12:40 (3. DS) **Location:** PC Lab SCH A 200a (Georg-Schumann Bau)

Topic assignment & group allocation

- **Date:** Tuesday, **April 21**, 2026, time: 09:20 – 10:50 (2. DS) **Location:** Online via Zoom

1st interim presentation

- **Date:** Tuesday, **May 19**, 2026, time: 09:20 – 12:40 (2. + 3. DS) **Location:** Online via Zoom

2nd interim presentation

- **Date:** Tuesday, **June 23**, 2026, time: 09:20 – 12:40 (2. + 3. DS) **Location:** Online via Zoom

Final presentation

- **Date:** Tuesday, **July 21**, 2026, time: 09:20 – 12:40 (2. + 3. DS) **Location:** Online (or in person upon request)

Objectives and requirements for all groups
Documentation and presentation

- Selection of suitable tools and programming frameworks for the implementation (e.g., Python, R, etc.)
- **Project management** to organize all tasks in a systematic manner within the group
- Meaningful documentation of the results in the form of a **written project report** (~ 20-35 pages for “usual” group size, longer series of screenshots and explanations can be organized in an appendix)
- **Interim presentations** (~ 12-15 min) should reveal the different ways of experimentation and the challenges the group is facing to have a vivid discussion with all groups (~ 5-8 min)
- The **final presentation** (~ 12-15 min) should reveal the most promising results, the applied practices to get there, and a lessons learned for an open discussion (~ 5-8 min)
- **Submission** of the project report, presentation files, scripts, visualizations, project files, literature sources, etc. via storage devices (e.g., USB stick, SD card) or file-sharing platforms

Objectives and requirements for all groups
Documentation and presentation

Project organization:

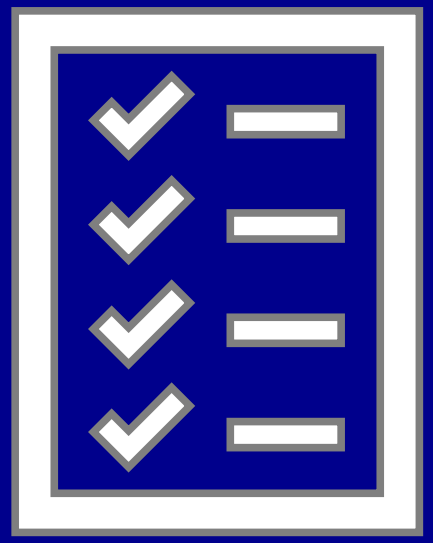
- Each groups will consist of ~3-4 students
- Communication via project lead to avoid redundant requests

Assessment:

- Project report (50%) + presentations (50%)
- Attendance is mandatory for all three (preliminary and final) presentations - for the entire slots(!)

Important dates

- First presentation of preliminary results: **May 19, 2026**
- Second presentation of preliminary results: **June 23, 2026**
- Submission deadline project report (via mail or file sharing): **July 17, 2026, 23:30**
- Final presentation: **July 21, 2026**



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It’s about you!



It’s about you!
You choose …

New hot topic (!!)

Agentic AI / multi-agent AI

- **… your group members and project organization**
  - Project management
  - Roles and team constellation
  - Task delegation and schedule
- **… your field of interest (recommended areas among others)**
  - Interpretable machine learning & explainable AI for tabular data
  - Natural language processing / text mining
  - Computer vision / image processing
- **… your dataset**
  - Option 1: “**Bring your own topic & dataset**” (e.g., public data sources, social media, commercial platforms, home applications, company partners, etc.)
  - Option 2: **Choose a topic from our selection** based on public repositories and benchmark settings
- **… your set of methods**
  - Core set of “expected” methods and tools
  - Additional methods for optional tasks

Choice of dataset: Option 1
“Bring your own topic & dataset”

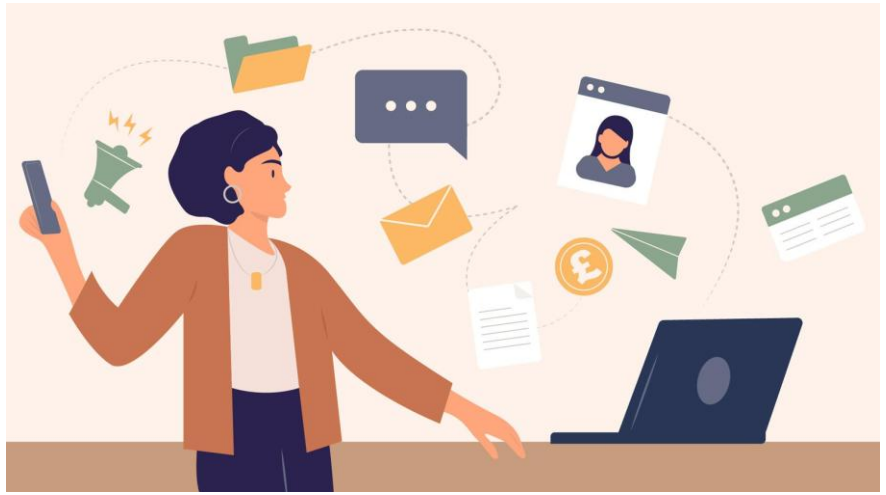
You are welcome to contribute your own topic/data from your private or professional setting which cannot be captured from publicly available sources/assets.

Private settings

- Server logs, smart home protocols, shopping behavior, smart watch activities, social media activities, …

Company settings

- Web content, manuals, customer complaints, company reports, process documentation, …

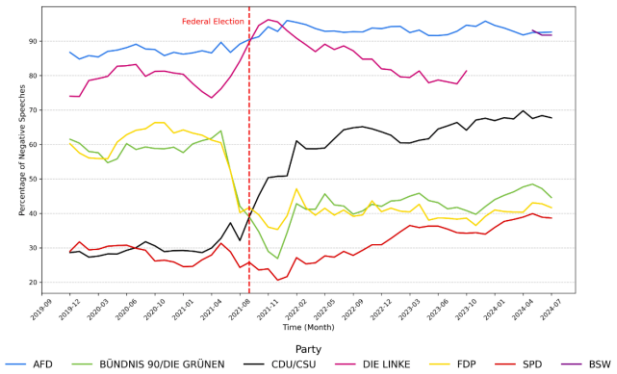


Make sure that **you are allowed to share and process the data** without any breaches of confidentiality.

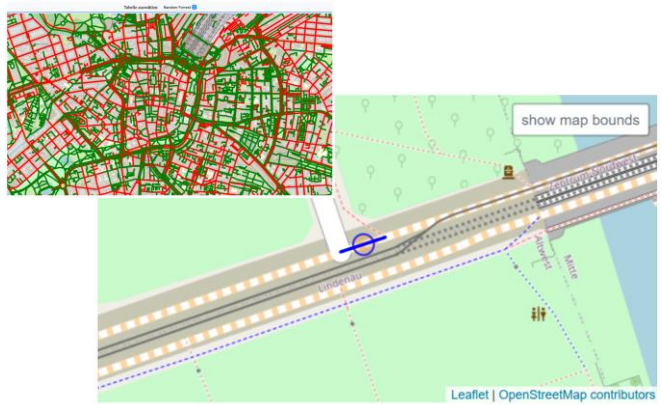
Please submit a short proposal (1-2 pages) with relevant information within the next few days (e.g., rough data properties, possible tasks, possible methods, expected outcome, etc.)

Choice of dataset: Option 1

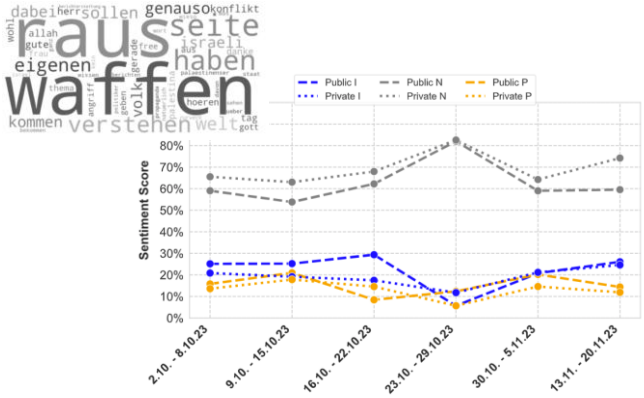
"Bring your own topic & dataset" – possible topics of interest



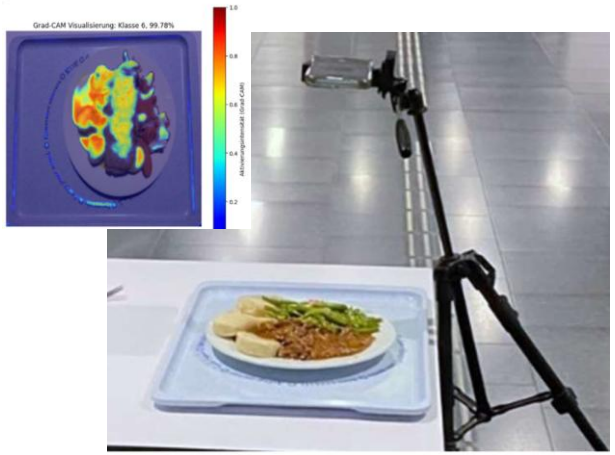
Parliament speech analysis



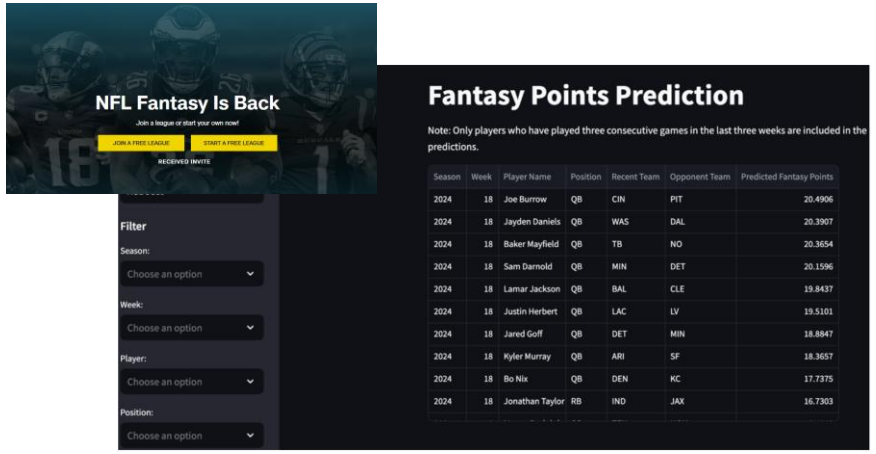
Bike accident prediction



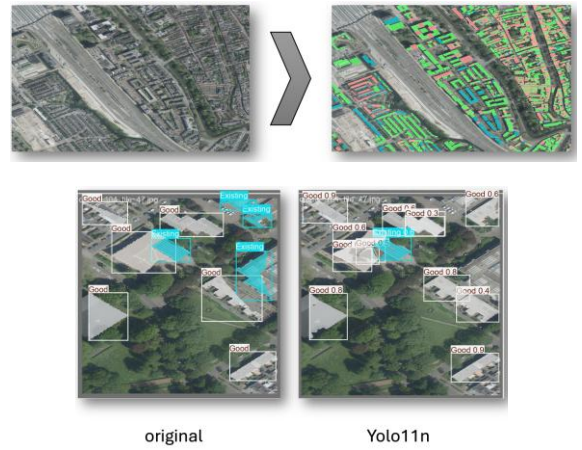
Hate speech detection



Automated meal recognition



Fantasy football point prediction



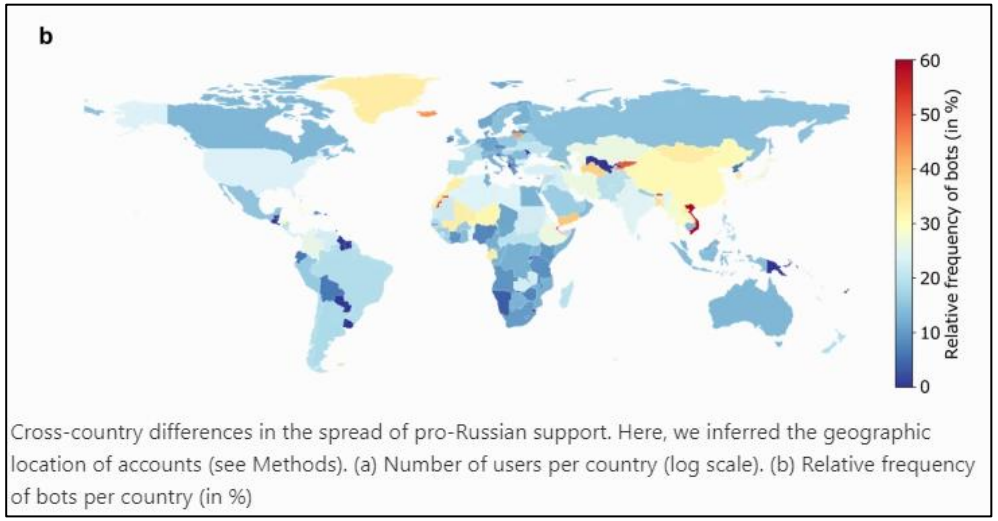
Solar roof detection

Choice of dataset: Option 1

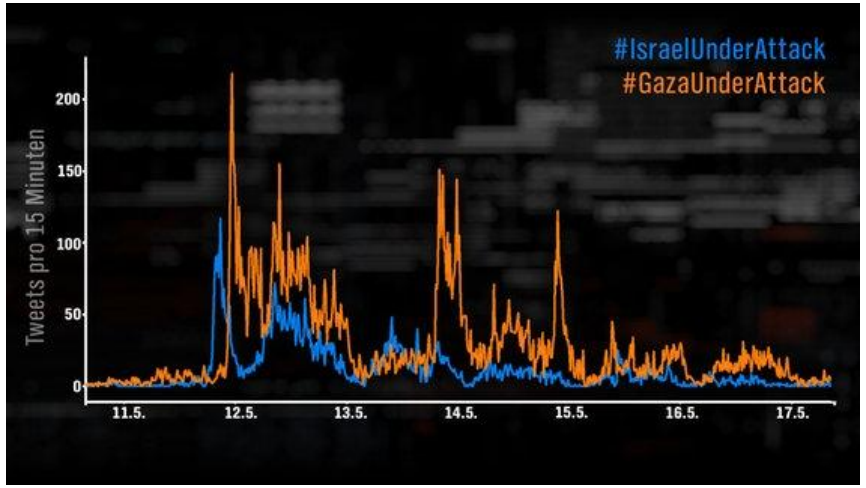
“Bring your own topic & dataset” – possible topics of interest

Politically charged topics in social media (e.g., Twitter)

- **#ukrainewar, #istandwithrussia, #istandwithputin**
- E.g., *Geissler, D., Bär, D., Pröllochs, N. et al. Russian propaganda on social media during the 2022 invasion of Ukraine. EPJ Data Sci. 12, 35 (2023).* *https://doi.org/10.1140/epjds/s13688-023-00414-5*



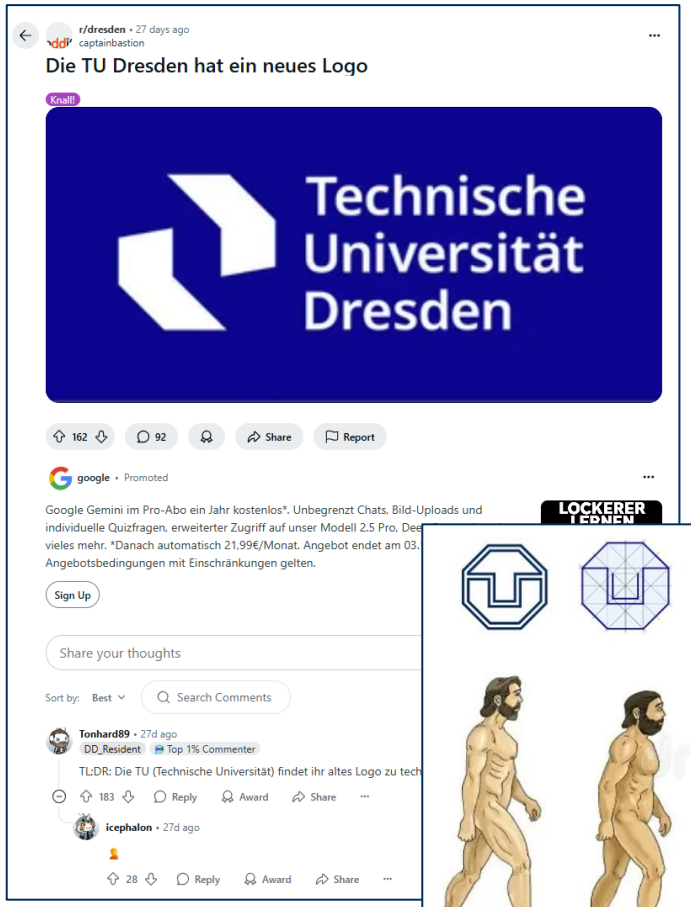
Cross-country differences in the spread of pro-Russian support. Here, we inferred the geographic location of accounts (see Methods). (a) Number of users per country (log scale). (b) Relative frequency of bots per country (in %)



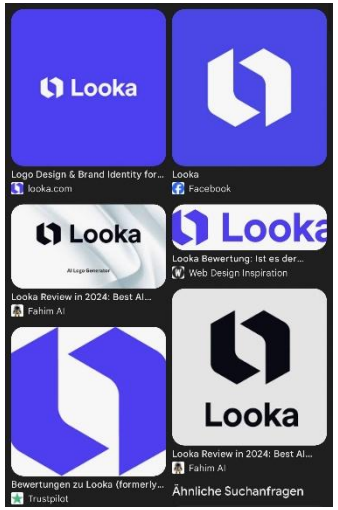
- **#IsraelUnderAttack vs. #GazaUnderAttack**
- https://www.ndr.de/fernsehen/sendungen/zapp/Der-Krieg-um-die-Hashtags,nahostkonflikt156.html

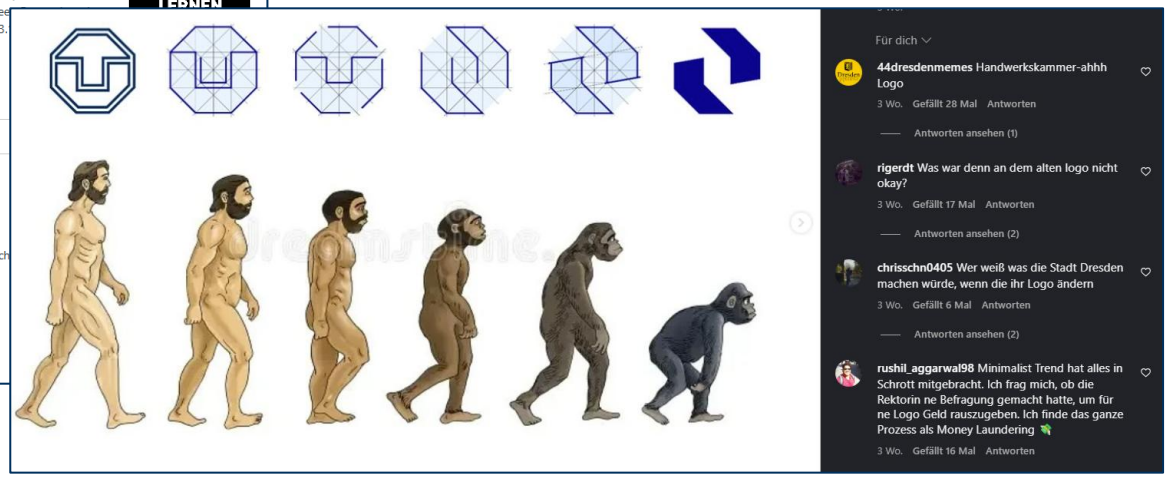
Choice of dataset: Option 1

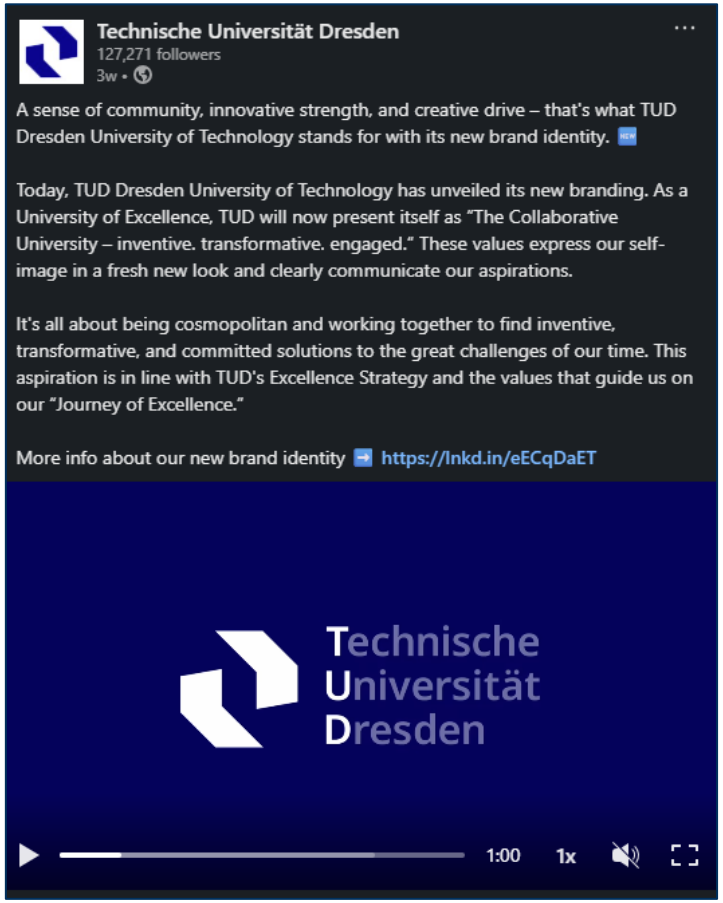
"Bring your own topic & dataset" – possible topics of interest



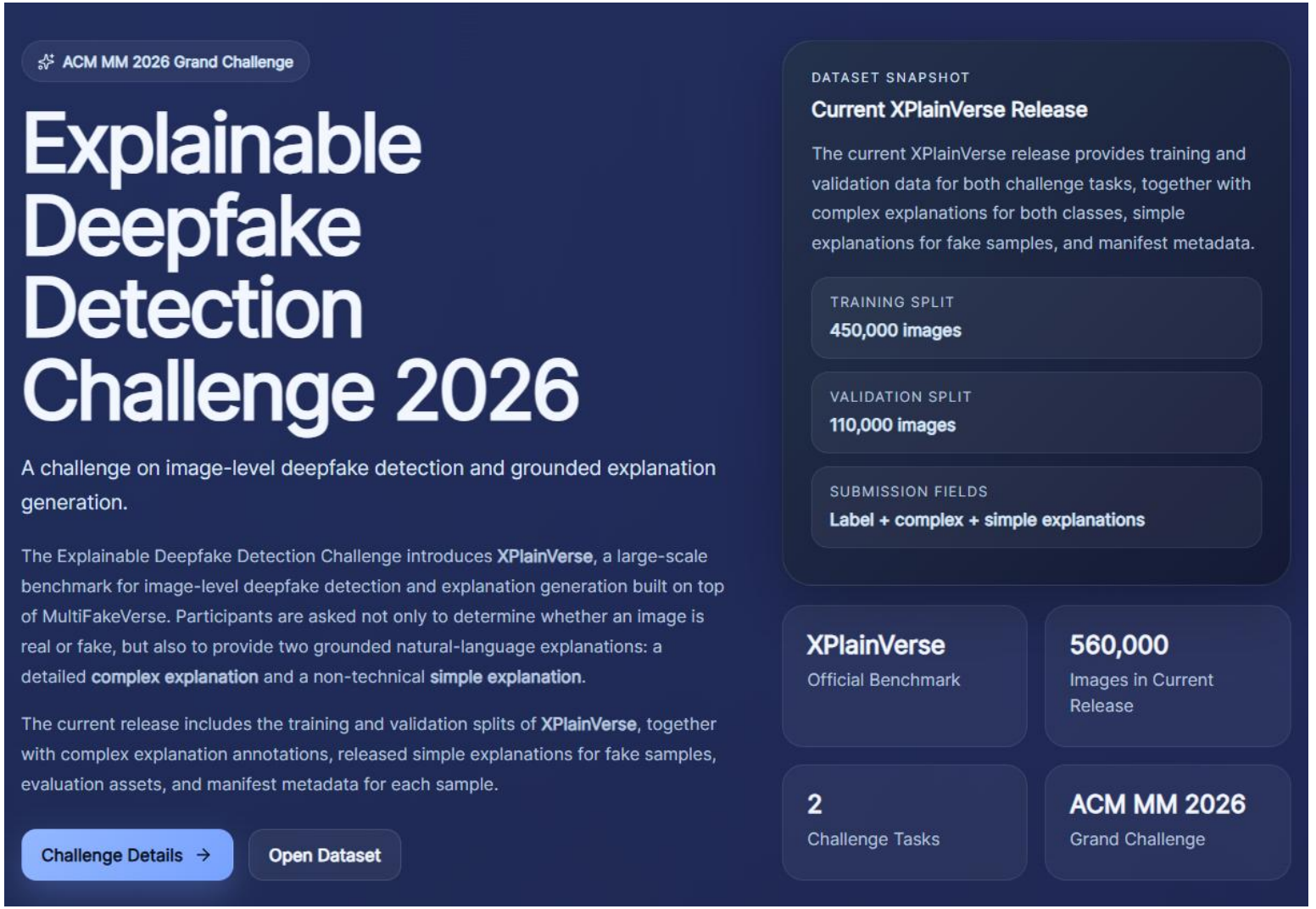








Additional topic: https://explainable-deepfake-detection.github.io/



Choice of dataset: Option 2

Choose a topic from our selection

- **Topic 1:** Cycling stage prediction with interpretable ML
- **Topic 2:** Wearable health data & interpretable ML
- **Topic 3:** LLM-as-a-Judge — Evaluating AI evaluation
- **Topic 4:** Multimodal product understanding in e-commerce
- **Topic 5:** Agentic AI pipeline
- **Topic 6:** Rental price prediction with interpretable ML
- **Topic 7:** Generalizability of intensive care predictions

Topic 1: Cycling stage prediction with interpretable ML

Can we explain why a breakaway wins? A sports analytics take on IML

Motivation & tasks

- Pro cycling has rich historical data, yet existing ML work is almost entirely black-box
- **Task ideas:** Stage outcome prediction, rider performance ranking, breakaway success classification, and more
- **Core:** Engineer features from stage profiles, elevation, rider form, head-to-head records, and weather API data
- **Core:** Compare inherently interpretable models (e.g., IGANN, EBM) vs. gradient boosting (e.g., XGBoost, LightGBM) vs. TabPFN; use interpretable model outputs as the primary explanation
- **Optional:** Post-hoc explanation comparison (SHAP) for gradient boosting and TabPFN; learn-to-rank framing and benchmark against published results; web app development/deployment

Data & readings

- **LeTourDataSet** (GitHub): Every TdF stage 1903–2025, ready-to-use CSV
- **cyclingdata** R package: 11,125 stages across all Grand Tours and major stage races
- **procyclingstats** Python package: Rider-level features, results, and rankings
- Kholkine et al. (2020). A Machine Learning Approach for Road Cycling Race Performance Prediction. MLSA.
- Sprangers et al. (2021). A Learn-to-Rank Approach for Predicting Road Cycling Race Outcomes. Frontiers in Sports.

https://www.letour.fr/de, https://github.com/thomascamminady/LeTourDataSet

Topic 2: Wearable health data & interpretable ML

What does your Garmin know about your stress? And can the model explain it?

Motivation & tasks

- Wearables capture physiological signals relevant to health, but predictions without explanations are hard to trust in this context
- **Task ideas:** Stress detection, sleep stage classification, activity recognition, and more
- **Core:** Extract tabular features from raw physiological time series using NeuroKit2 (HRV metrics, EDA features, skin temperature statistics)
- **Core:** compare inherently interpretable models (e.g., IGANN, EBM) vs. gradient boosting vs. TabPFN; use interpretable model outputs as the primary explanation
- **Optional:** Post-hoc explanation comparison (SHAP) for gradient boosting and TabPFN; bring your own Garmin, Oura, or Apple Watch data; cross-individual generalization analysis

Data & readings

- **DREAMT (PhysioNet 2024):** Sleep stage estimation, multi-sensor wearable + polysomnography, open access
- **Wearable Stress Dataset (PhysioNet 2025):** Stress/exercise classification, Empatica E4, no credentialing required
- Optionally: Garmin Connect export, Oura API, or Apple Health data
- Wang et al. (2024). DREAMT. PhysioNet / PMLR 248.
- Hongn et al. (2025). Wearable Device Dataset from Induced Stress and Structured Exercise Sessions. PhysioNet.

Topic 3: LLM-as-a-Judge — Evaluating AI evaluation

Are LLMs reliable judges? A systematic bias audit with real consequences

Motivation & tasks

- LLMs are increasingly used as automated evaluators but carry systematic biases: Verbosity (longer responses rated higher), position (first option preferred), and self-preference (own model family favored)
- **Task ideas:** Bias audit across judge models, domain-specific calibration analysis, human vs. LLM agreement study, and more
- **Core:** Choose one domain, prompt at least 2 judge models, quantify at least 2 bias types statistically, and compare against a partial human ground truth
- **Core:** The dataset is generated by your group through structured prompting protocols
- **Optional:** German-language evaluation task, mitigation strategies (chain-of-thought, calibration prompts), additional judge models or domains

Data & readings

- **Student-generated** evaluation data via structured prompting protocols
- **MT-Bench, FLASK, BiGGen Bench:** Established reference datasets with human-annotated ground truth
- Zheng et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. NeurIPS.
- Chen et al. (2024). Humans or LLMs as the Judge? A Study on Judgement Bias. EMNLP.
- Jiang et al. (2024). A Survey on LLM-as-a-Judge. arXiv:2411.15594.
- Kim et al. (2024). BiGGen Bench. arXiv:2406.05761.

Topic 4: Multimodal product understanding in e-commerce

Text, image, or price: which modality actually drives the prediction?

Motivation & tasks

- Product listings combine text and structured metadata — most models cannot explain which modality drives a prediction
- **Task ideas:** Product quality classification, review helpfulness prediction, fake review detection, and more
- **Core:** Build unimodal baselines for text and tabular modalities separately
- **Core:** For the tabular modality, compare inherently interpretable models vs. gradient boosting vs. TabPFN
- **Core:** Implement text + tabular fusion and attribute predictions across modalities
- **Optional:** SHAP for gradient boosting and TabPFN; image modality; web app development/deployment

Data & readings

- **Amazon Reviews 2023** (McAuley Lab / HuggingFace): Reviews, ratings, text, images and metadata through Sep 2023
- **EMMa dataset:** 2.8M Amazon product listings with images, text, price, ratings, and material annotations, designed for multi-task learning
- Hou et al. (2024). Bridging Language and Items for Retrieval and Recommendation. arXiv:2403.03952.
- Lütkenhaus et al. (2022). EMMa: An Extensible Multi-modal Multi-task Object Dataset. NeurIPS Datasets Track.

Topic 5: Agentic AI pipeline (open topic)

Design, build, and evaluate a multi-agent AI system for a task of your choice

Motivation & tasks

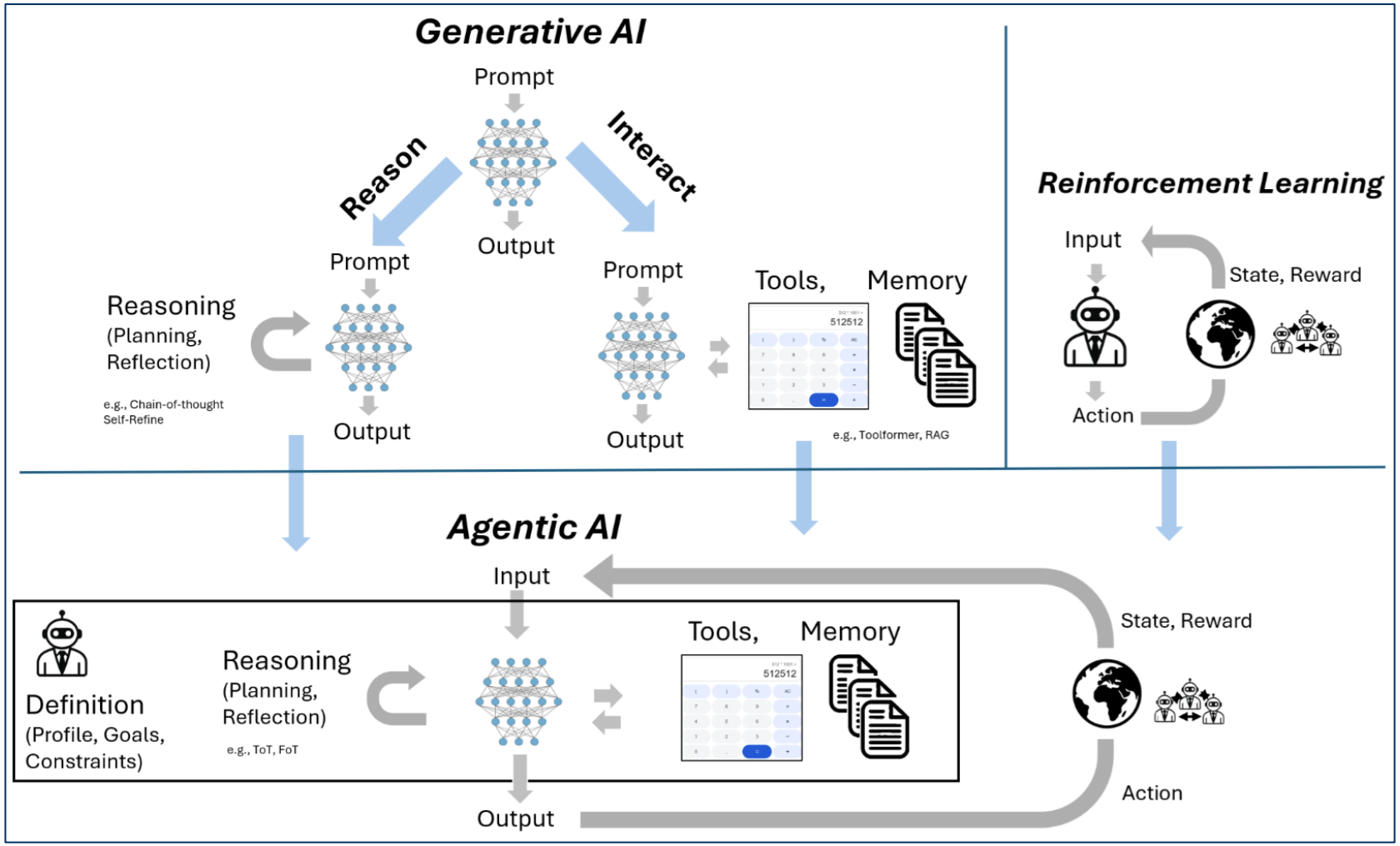
- Agentic AI systems that autonomously plan, use tools, and coordinate across sub-tasks represent the current frontier of AI application design in IS
- **Task ideas:** Literature screening, automated data analysis, document processing, research support, and more
- **Open topic:** Groups propose their own domain and submit 1-2 pages proposal for approval by **April 20**
- **Proposal must specify:** Application domain, bounded task definition, measurable success criteria, and a concrete non-agentic baseline
- **Optional:** User study on trust and usefulness, failure mode analysis, multi-agent verification pattern

Frameworks & readings

- **LangGraph / AutoGen / CrewAI:** Recommended frameworks for implementation
- Schneider (2025). Generative to Agentic AI: Survey, Conceptualization, and Challenges. arXiv:2504.18875.
- Allmendinger et al. (2026). Multi-Agent AI Systems. Electronic Markets / Springer.
- Feuerriegel, Hartmann, Janiesch & Zschech (2024). Generative AI. BISE.
- Wang et al. (2023). A Survey on Large Language Model Based Autonomous Agents. arXiv:2308.11432.

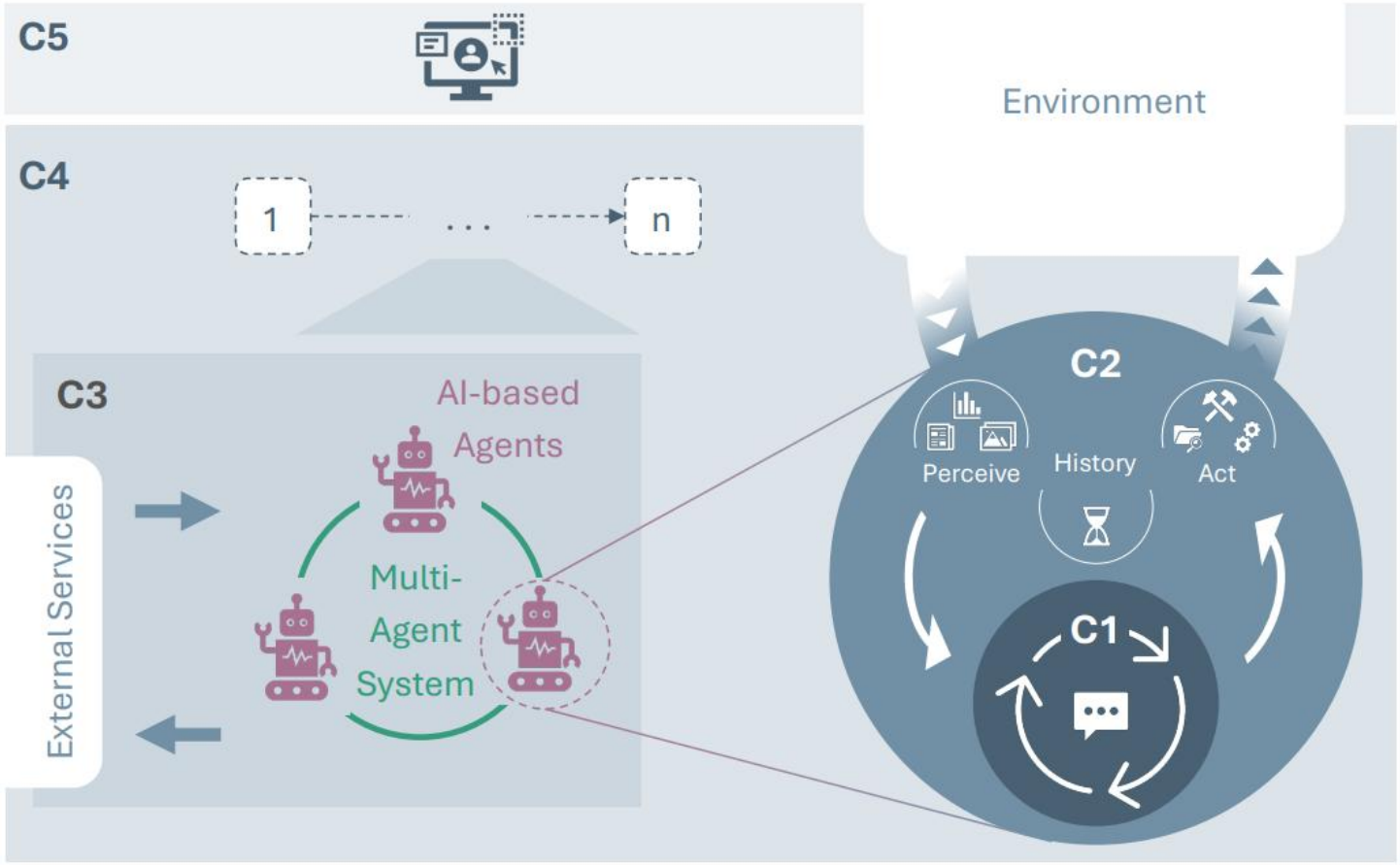
Agentic AI / multi-agent AI

Schneider (2025): https://arxiv.org/pdf/2504.18875



Agentic AI / multi-agent AI

Allmendinger et al. (2026): https://link.springer.com/article/10.1007/s12525-025-00862-z



C5 | Interaction Interface

User interface (UI) or system integration, including application programming interfaces (APIs).

C4 | Agent-integrated Workflow

Configurable arrangement of agents and their interactions, defining task sequences, dependencies, and termination.

Multi-Agent System

AI-based Agent

C3 | Dynamic Orchestration

Conducted composition of and decision making between agents and external services.

C2 | Data-centric Perception & Action

Individual perception and action enable agents to process data and act adaptively in the environment.

C1 | Foundation Model

(Large) AI models enable multimodal input processing, reasoning, and output generation.

Topic 6: Rental price prediction with interpretable ML

Beyond coefficients: What can interpretable ML reveal about German rental markets?

Motivation & tasks

- Linear models predict rental prices well — but GAMs could reveal nonlinear relationships and spatial patterns that coefficients cannot capture
- **Task ideas:** Rental price prediction, price driver analysis, regional comparison, temporal trends, and more
- **Core:** Engineer spatial features from georeferenced listings (distance to center, public transport, amenities)
- **Core:** Compare inherently interpretable models (e.g., IGANN, EBM) vs. gradient boosting vs. TabPFN; use interpretable models as the primary explanation
- **Optional:** SHAP for gradient boosting and TabPFN; city-level model comparison; web app; …

Data & readings

- **RWI-GEO-RED (FDZ Ruhr):** ImmoScout24 listings 2007–2024, ~50% of German online market, georeferenced — scientific access, application required
- **Apify ImmoScout24 Scraper / Immospider** (GitHub): Scraping alternative for immediate access — listing prices, not actual transaction prices
- Lorenz et al. (2022). Interpretable Machine Learning for Real Estate Market Analysis. Real Estate Economics.

Topic 7: Generalizability of intensive care predictions

How well can ML models generalize between different hospitals?

Motivation & tasks

- Machine learning models for intensive care units are often trained on data from a single hospital, which can limit their performance in other hospitals due to differences in clinical practices and data collection
- **Task:** Predict mortality, length of stay and readmissions with existing pipeline
- **Core:** Integrate intensive care data from different hospitals in the US (eICU)
- **Core:** Extract and preprocess data from multiple intensive care units and integrate them into the existing pipeline
- **Core:** Integrate additional machine learning models
- **Optional:** Gain access to and integrate the HiRID and SIDdb datasets.

Data & readings

- **eICU:** Multi-center intensive care dataset with over 200,000 patient admissions from more than 200 hospitals in the US
- **HiRID Dataset:** High-resolution intensive care unit patient data from Bern University Hospital
- **SICdb Dataset:** Intensive care dataset with over 27,000 admissions from University Hospital Salzburg
- Bohlen et. al (2026). Toward generalizable and interpretable machine learning models in healthcare: Insights from ICU outcome predictions. HCMS.

Overview of expected steps and methods

1) **Data collection, integration, and cleaning** (e.g., filtering, harmonization, noise removal, removal of duplicates, missing values, feature scaling, etc.)
2) **Descriptive data analysis for data exploration & data understanding** (e.g., dataset properties, distribution, univariate, bivariate, multi-variate, correlations, imbalance, etc.)
3) **Development of an automated machine learning pipeline** (feature preparation, sampling/data splitting, model training, evaluation strategy, etc.)
4) **Implementation of different machine learning models**
   - **Simple models** (e.g., *linear models, decision tree*)
   - **Advanced flexible models** (e.g., *tree-based ensembles, deep neural networks, tabular foundation models*)
   - **Advanced interpretable models** (e.g., *generalized additive models*)
5) **Model evaluation, diagnosis, interpretation** (post-hoc explanations, interpretable model structure)

Additional steps and methods

- **Collection/integration of additional datasets** (e.g., to improve results, to validate external generalizability)
- **Experimentation and systematic comparison** of different techniques/models/approaches (e.g., different data preparation steps and/or ML models to assess their impact on prediction results)
- **Development of an advanced (web-based) software application** for the deployment of the models/results (e.g., interactive user interface, dashboards, conversational assistance, etc.)
- **Comparison of machine learning results with human baseline performance**

The central question…

After getting this short introduction …

WHO WILL STAY ON BOARD?

Please do not hesitate to contact me!

Thank you very much for your attention!

Any questions?

Patrick Zschech
patrick.zschech@tu-dresden.de

TU Dresden
Professorship for Intelligent Systems and Services



Recommended readings

- Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business & Information Systems Engineering*, *66*(1), 111–126. https://doi.org/10.1007/s12599-023-00834-7
- Janiesch, C., Zschech, P., & Heinrich, K. (2021). Machine learning and deep learning. *Electronic Markets*, *31*(3), 685–695. https://doi.org/10.1007/s12525-021-00475-2
- Zschech, P., Weinzierl, S., & Kraus, M. (2025). Inherently Interpretable Machine Learning: A Contrasting Paradigm to Post-hoc Explainable AI. *Business & Information Systems Engineering*. https://doi.org/10.1007/s12599-025-00964-0